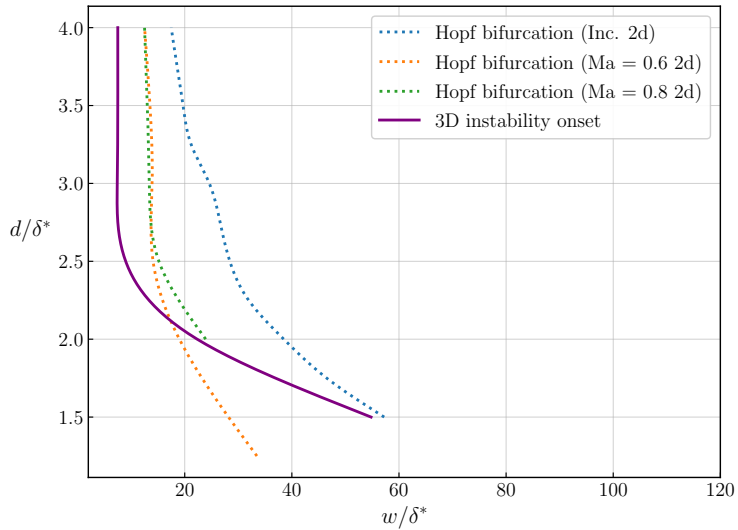


IMPERIAL

Stability diagram

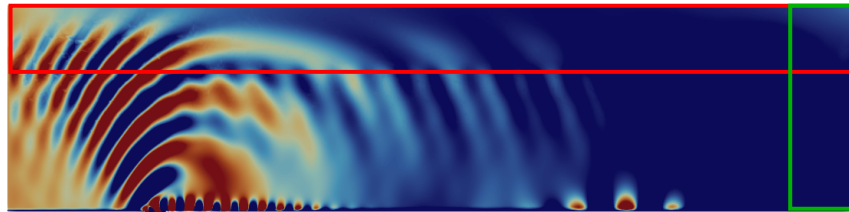
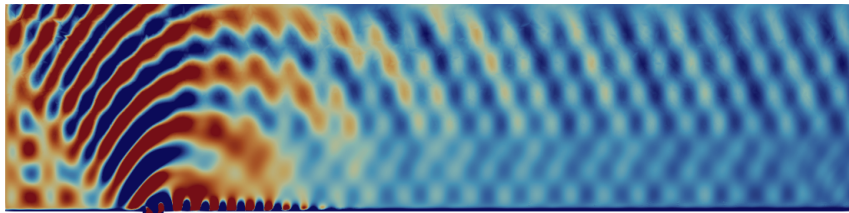
Víctor Ballester
December 3, 2025

Stability diagram



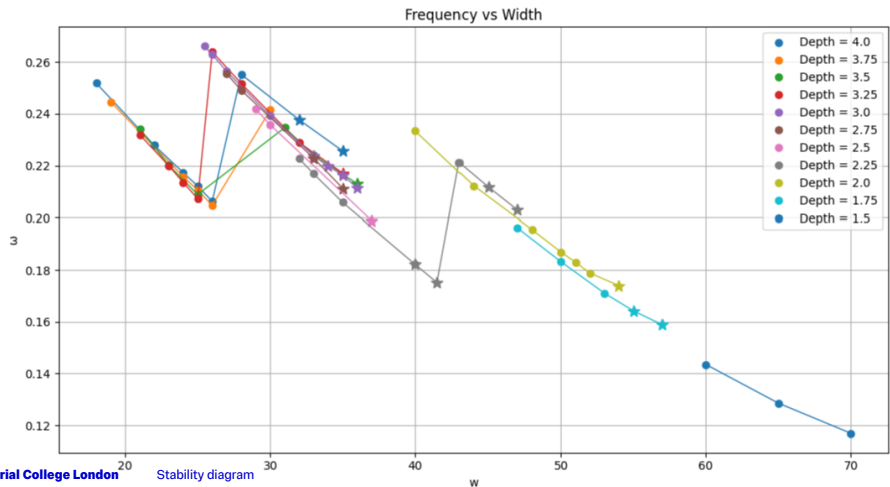
Sponge region

- With and without sponge region ($d = 3$, $w = 15$). Top and outflow boundary conditions. Inflow no (so far) because of stability issues (bad resolution of the boundary layer with $p=1$ near the BC)

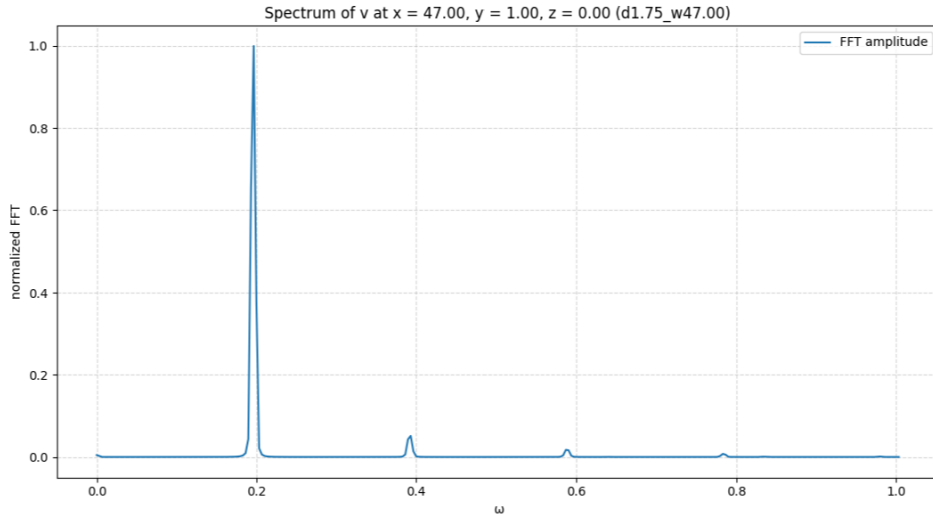


2nd bif Inc

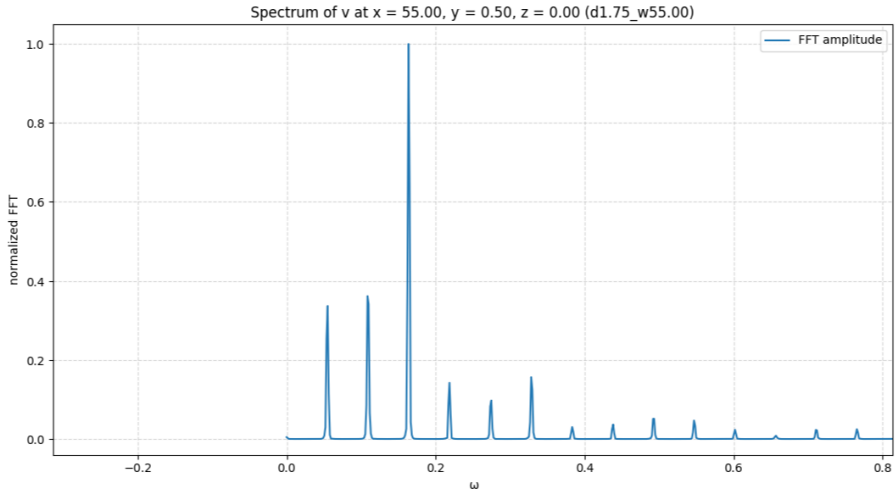
- Most energetic frequency as function of w (in the linearly unstable regime).



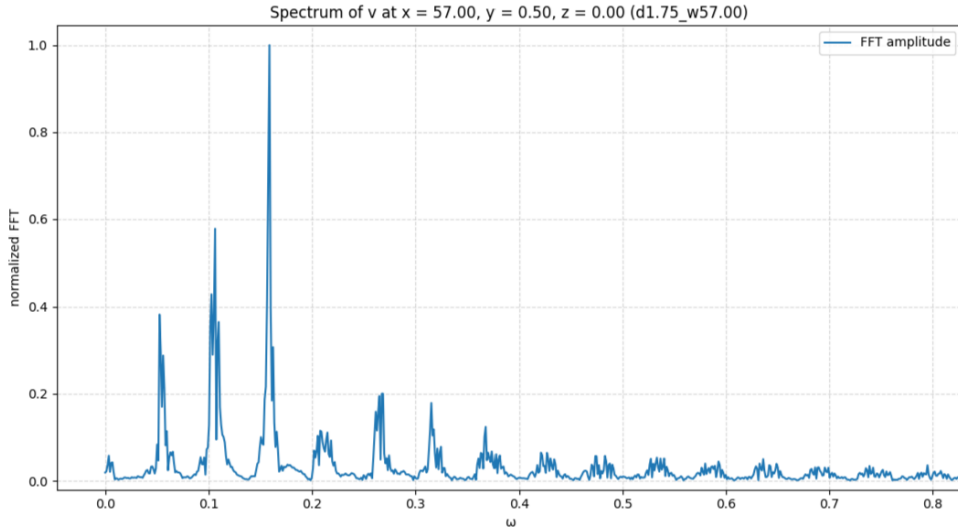
Example of $d = 1.75 - 1$



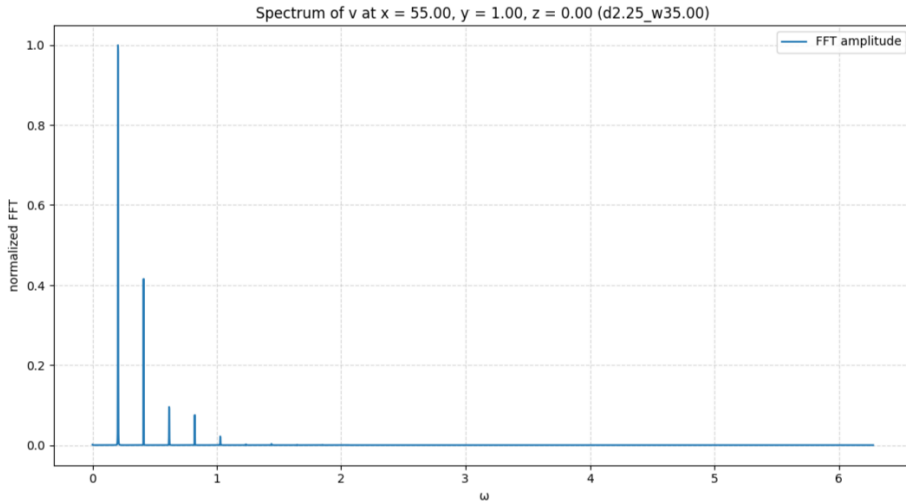
Example of $d = 1.75 - 2$



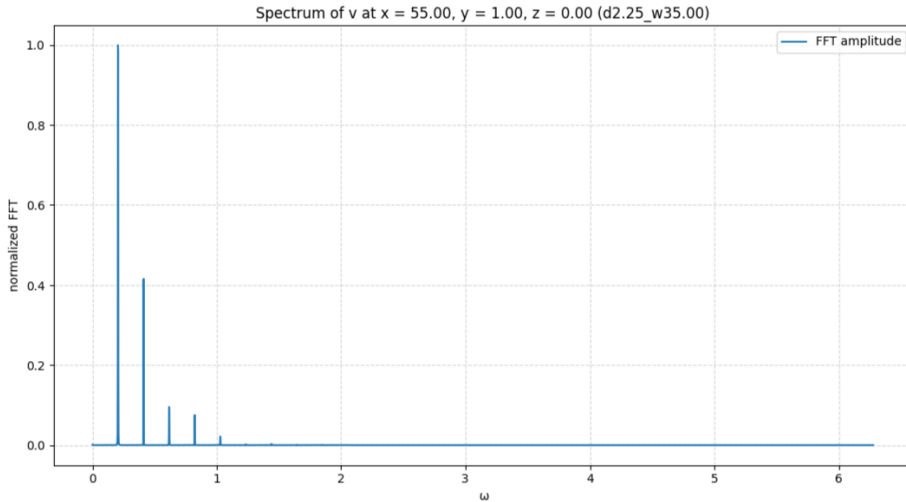
Example of $d = 1.75 - 3$



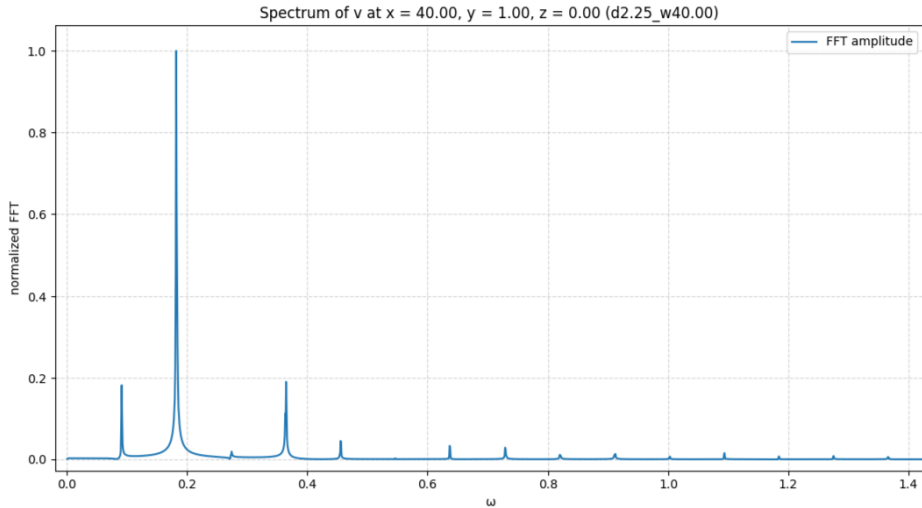
Example of $d = 2.25 - 1$



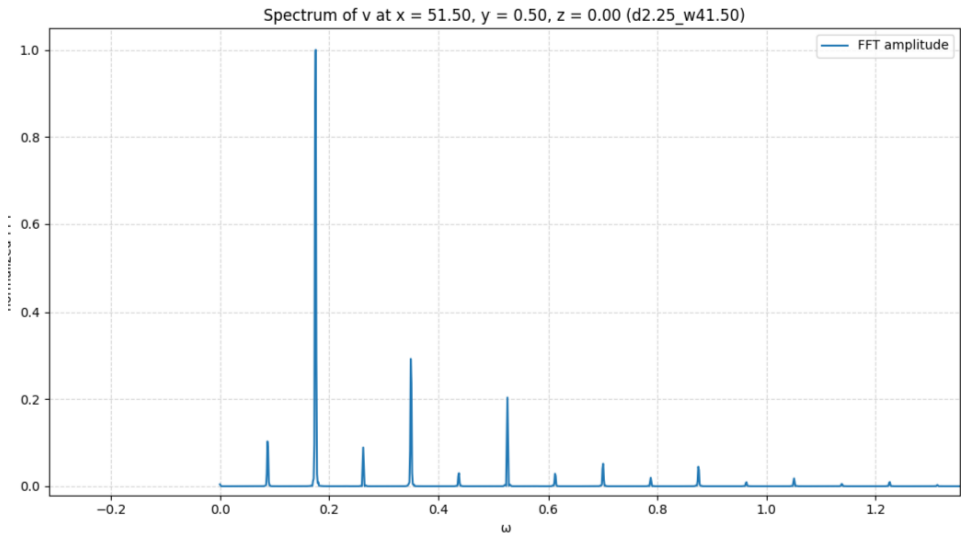
Example of $d = 2.25 - 1$



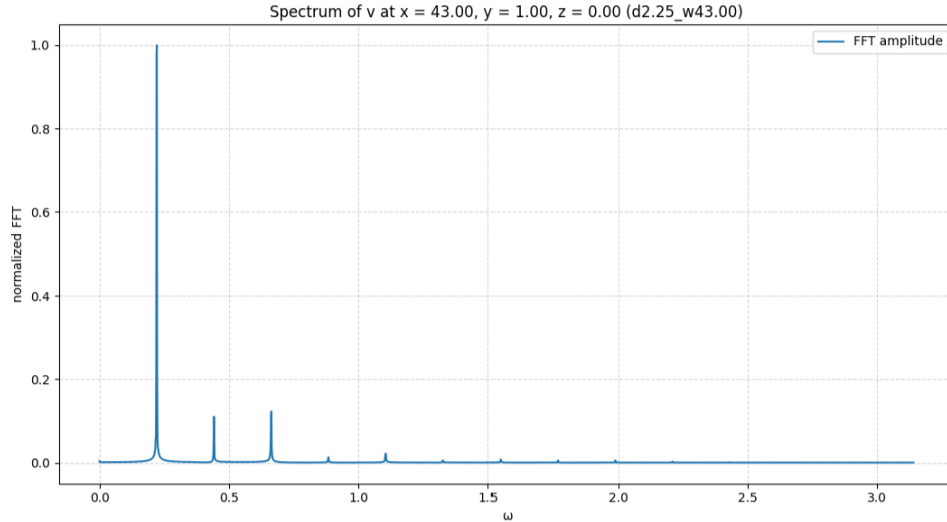
Example of $d = 2.25 - 2$



Example of $d = 2.25 - 3$



Example of $d = 2.25 - 4$



Example of $d = 2.25 - 5$

